Emergency Severity Index (ESI) Classification: Baseline Model Results

CariSurg MedTech Pathways - Week 6 Final Deliverable - Mercer General Hospital

Purpose

This report evaluates two baseline machine learning (ML) models, logistic regression and decision tree, which were trained to predict ESI levels from triage data. The objective at this stage is not to produce a deployment ready model but to determine whether routine triage contains enough information to support a future AI-assisted triage decision support tool and identify the current limitations

Data & Methods

- **Dataset:** A cleaned triage dataset containing 55,121 emergency department (ED) encounters.
- **Train & Test Split:** The dataset was divided into 80% training data (44096 encounters) and 20% testing data (11024 encounters). The split was stratified which means the dataset contained similar proportions of patients from each ESI category. This ensures that there is a fair evaluation of model performance.
- **Features:** The model used patient vital signs (such as heart rate and blood pressure) and chief complaints that were recorded during triage. Demographics, administrative and patient outcome information (e.g., disposition and previous disposition) was excluded to prevent the models from learning information that would not be available during triage assessment.
- **Models Compared:** Three models were evaluated. A simple baseline model was first used as a reference point to show which performance could be achieved without any advanced prediction methods. A logistic regression model was used on adjusted data and the decision tree model was tested using original data. The performance of each model was compared to determine which approach worked best.

Results

Model Performance Comparison

Three models were evaluated to determine how well they could predict patient acuity levels (ESI Level 1-5). The dummy classifier was used for a baseline comparison, while logistic regression and decision tree models were tested to see if they could improve predictive performance.

Model	**Accuracy**	**Overall Performance (Macro F1)**	**Ability to Identify Most Urgent Cases (ESI Level 1 Recall)**
Dummy (baseline)	0.375	0.204	0.00
Logistic Regression	0.667	0.495	0.25
Decision Tree	0.556	0.216	0.00

Dummy Classifier: The baseline model achieved a 37.5% accuracy but failed to identify urgent patients (ESI level 1).

Logistic Regression: This model performed the best as it achieved an accuracy of 66.7% as well as a high Macro F1 score (49.5%). However, it was only able to identify 25% of ESI level 1 cases meaning some high acuity patients were missed.

Decision Tree: The decision tree performed better than the dummy classifier but worse than the logistic regression. It also failed to identify ESI level 1 cases.

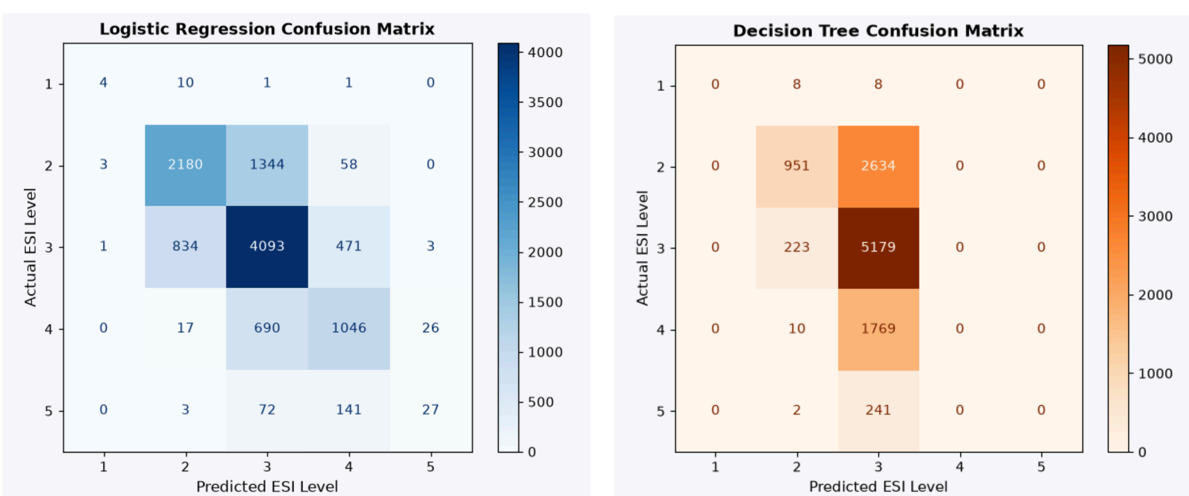
<u>Logistic Regression Per-Class Performance:</u>

ESI	Precision	Recall	F1-score
1 (most urgent)	0.500	0.250	0.333
2	0.716	0.608	0.658
3	0.660	0.758	0.706
4	0.609	0.588	0.598
5 (less urgent)	0.482	0.111	0.181

Note: The F1-score represents the balance between precision and recall. It was calculated using the following formula:

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

ESI Level 1 (Most Urgent): The model struggled with high risk cases, only identifying 25% of true ESI level 1 patients. This is crucial as missing urgent patients can result in serious clinical consequences.
ESI Level 2: The model performed well as it correctly identified most ESI level 2 cases with a good balance between prediction and detection.
ESI Level 3: This category performed the best, with the highest F1 score of 0.706. This shows that the model was effective at identifying ESI level 3 patients
ESI Level 4: The model showed moderate performance, correctly identifying over half of ESI level 4 cases.
ESI Level 5 (Least Urgent): The model also struggled with this category as it only detected only a small portion of true ESI level 5 cases.



Left: Logistic Regression. Right: Decision Tree. Rows = actual ESI level, columns = predicted. The tree's predictions cluster almost entirely in the ESI 3 column regardless of the true class.

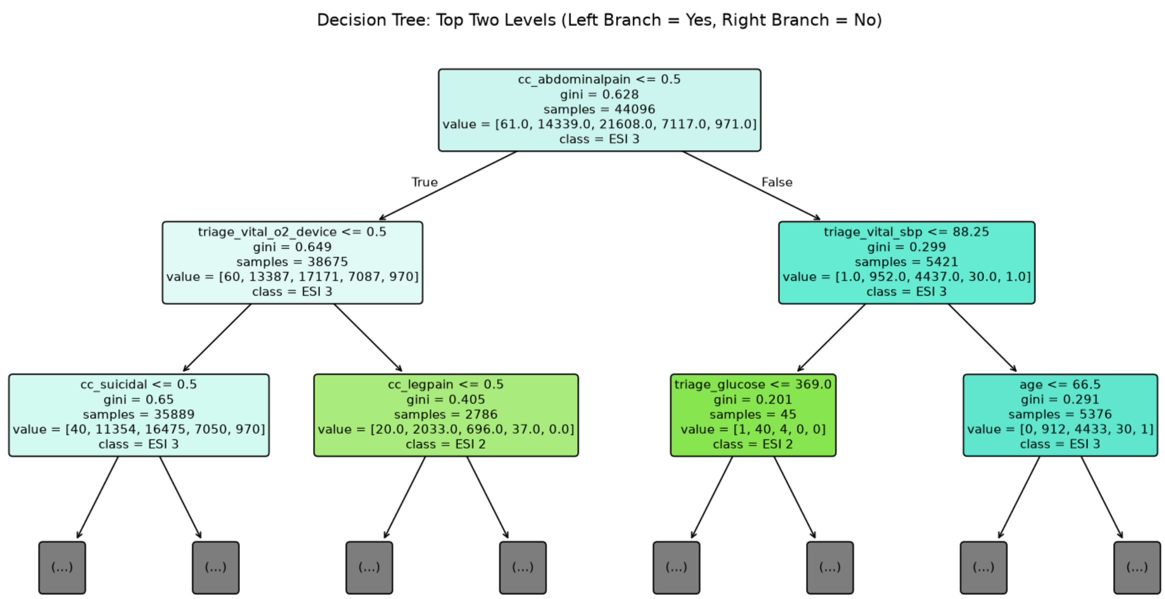
Metric Justification

Accuracy alone is not enough for this dataset because the ESI classes are unevenly distributed. Most patients were assigned ESI level 2-4, while ES1 level 1 and 5 were less common. A machine learning model could achieve high accuracy as it mostly predicts the common classes while missing rare cases.

Macro F1 was used as it gives an equal importance to every ESI level. This helps show when the model performs poorly with less common but clinically important cases. ESI level 1 recall was measured separately as missing urgent patients is a serious error a triage model can make.

Failure Mode Reflection

Decision Tree: this method mostly predicted levels that were more common (ESI level 2-4) and failed to identify ESI level 1 patients achieving a 0% recall. Its higher training accuracy compared with testing suggests it is learning the data too closely rather than general patterns.



The first two decision points from the depth-12 tree are shown to explain how the model makes predictions. The model first splits the patients by abdominal pain but after two levels it mostly predicts ESI level 3, showing it favours the most common class

Logistic Regression: Logistic regression performed better by identifying patients across all ESI levels. However, it only detected 25% of ESI level 1 cases, meaning it missed most of the urgent patients. The model also struggled with ESI level 5 patients.

Clinical Implication: Neither model is reliable enough to make independent triage decisions. Since both models miss most high-risk patients, it should support clinicians by highlighting patterns or risks rather than replacing clinical judgement.